MINING DEALER BEST PRACTICE

Attachment for the Removal & Install of the Steering Accumulators of 797F Trucks

Maintenance and Repair | Application | **Component Life Management** | Component Renewal (CRC) | MARC Management

September 2016
3.04-160921-01

1.0 Introduction

Mining equipment are an important part of the production process of any mining operation, and are critical in meeting the production goals. To ensure the necessary availability and reliability of the equipment, it is imperative to include precise maintenance and repair tasks; tasks that must be performed with safety, quality and efficiency.

When dealing with large size equipment such as the Caterpillar 797F trucks, the maintenance activities usually include tasks that require the handling of heavy large size components. This is the case in the Removal and Install of the 797F Steering Accumulator.

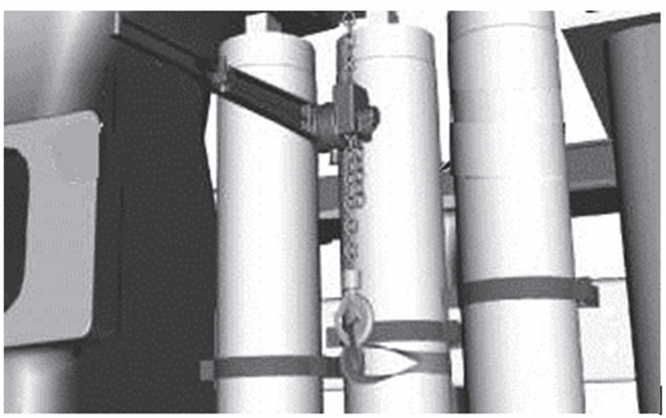
To achieve the objectives of safety, efficiency and quality, the maintenance department must ensure that the personnel is properly equipped with the necessary training, support information (technical and processes information), infrastructure, tools and support equipment.

This Best Practice refers to the design, manufacture, application and the related benefits of an attachment, developed to improve the Removal and Install of the steering accumulators of 797F trucks. This accessory was developed in line with the objectives of efficiency, quality and safety mentioned above.

2.0 Best Practice Description

2.1 Detection of the Opportunities

During the analysis of the R&I of the steering accumulators of the 797F trucks, we detected 4 important improvement opportunities. These were the starting point and the motivation to design and fabricate the attachment proposed in this Best Practice.



Areas of Opportunities

Safety

The traditional procedure recommended for the Removal & Install of the steering accumulators (Media # KENR8382-09), includes the use of slings/straps, lever puller tool and adequate hoist equipment. Due to the sensitive nature of the procedure, successful application relies on the experience of the technician(s) performing the task. The technician(s) must maintain the control and balance of the accumulator at all times to avoid the risk of damage to the part or equipment from striking the part during movement.

Any work that considers a “suspended” load is risky by nature. Special attention must be given to safety, especially with avoiding pinch points and avoiding contacting or striking the technician(s) with the swinging accumulator. Steering accumulator weight: 236 kg.

Time (Efficiency)

Based on our records and repair history, the average time required to remove and install a steering accumulator following the traditional recommended procedure is 7 hours machine down time. It is also important to mention that when working with a suspended load it is not recommended to perform any other work or tasks in parallel within the same working area.

Resources (Labor)

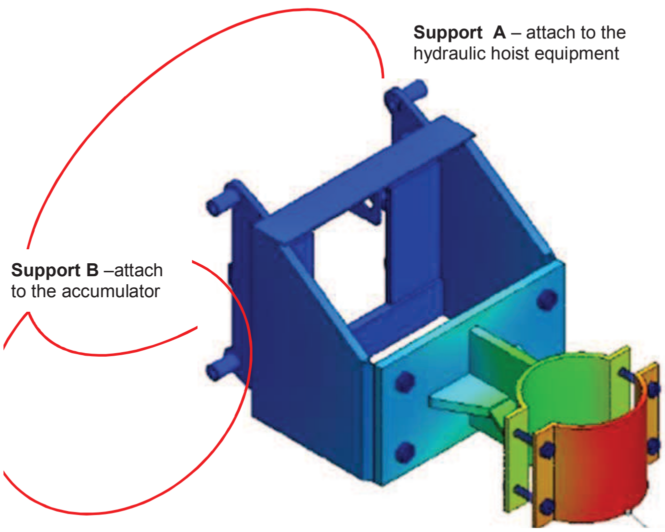
To complete the task, we normally assign 3 technicians, and record an average of 21 labor hours for completion of the task.

Quality

Some events of sub-standard quality have been observed and recorded during the R&I of the steering accumulators. These events have generally been attributed to the lack of control of the accumulator while it is being manipulated as a suspended load.

2.2 Proposed Improvement

The proposed solution / improvement contained in this BP, is the use of an attachment, that was specifically designed for the R&I of the accumulators (Image 1). The attachment is installed on the accumulator that will be manipulated, then a portable hydraulic hoist is connected to the attachment, such as the one showed in the image 2. From here on, we will identify this attachment as “ARIA-DMH”, which stands for **A**ttachment to **R**emove & **I**nstall the **A**ccumulators – Division Ministro Hales.



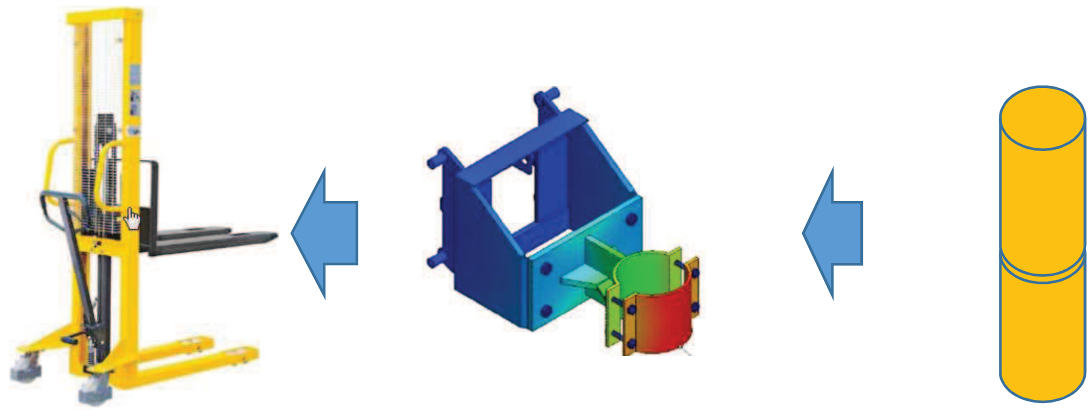
mage 1

The support A is connected to the hydraulic hoist equipment. And the support B is connected to the steering accumulator

Image 2

Type of hydraulic lift equipment selected.

It is important when selecting this equipment to observe the capacity, minimum 1000 kg and the vertical lift that must be at least 3 meters.



Hydraulic Hoist Equipment | **Attachment ARIA - DMH** | **Steering Accumulator**

The combination of hydraulic hoist equipment and a rigid attachment (ARIA-DMH) provides adequate control and maneuverability to perform the task with improved safety and efficiency. This proposed improvement changes the former procedure of handling a suspended load to a supported load, which is favorable in terms of safety.

The following pictures show the ARIA-DMH being used in our truck shop.

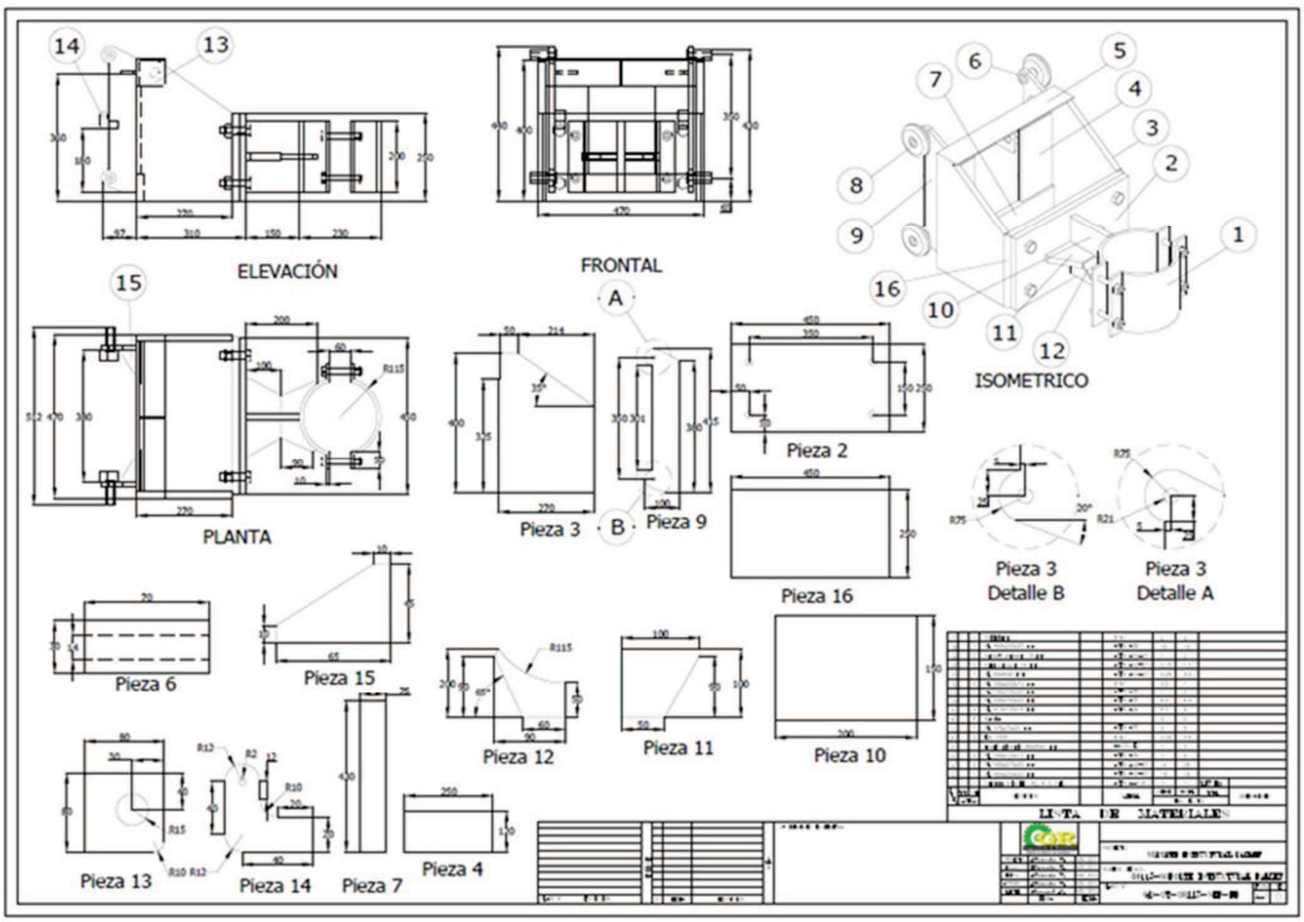
3.0 Implementation Steps

The following steps are recommended to follow while implementing this Best Practice:

3.1 Fabrication of the ARIA-DMH attachment
3.2 Selection and acquisition of the hydraulic lift as recommended
3.3 Technical certification of the ARIA-DMH attachment
3.4 Testing of the ARIA-DMH in a controlled environment
3.5 Final Adjustments
3.6 Documentation of the standard operating procedure (SOP)
3.7 Training of the technicians that will use the ARIA-DMH regularly (2 per shift at the minimum)
3.8 Release of the ARIA-DMH to work and start recording the benefits

3.1 Fabrication of the ARIA-DMH attachment

The ARIA-DMH was fabricated locally following the design specification shown in the attached drawing. The design and fabrication drawings were all made by Finning SA – DMH operation.



Note: We used a local company to fabricate the ARIA-DMH.

GR Ingeniería & Servicios
Av. Jerusalem No. 4166
Antofagasta, Chile
Jv.gringenieria@gmail.com

3.2 Selection and acquisition of the hydraulic lift recommended

Image 3

Hydraulic Lift selected

Brand:	NOBLELIFT
Model:	SFH1030-A
Capacity:	1000 kg
Vertical lift:	3 meters

Acquire the hydraulic lift recommended. Observe the specifications of capacity and vertical reach.

3.3 Technical certification of the ARIA-DMH

Once the attachment is fabricated, it is important to run it through a technical inspection / certification to ensure the quality of the welding and the structural capacity.

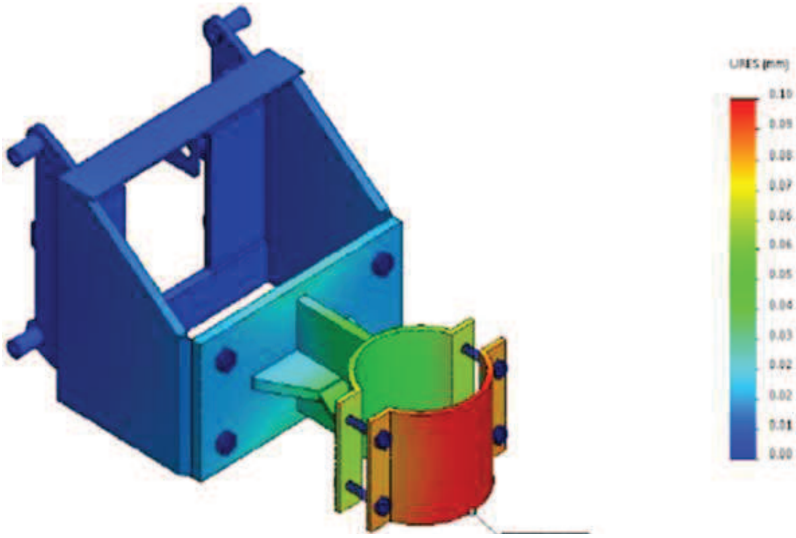
08115- ACUMULADOR DE DIRECCION

GR Ingeniería & Servicios
Solucion de calidad a sus necesidades..!

MEMORIA DE CALCULO PARA HERRAMIENTA DE INSTALACION DE ACUMULADOR DE DIRECCION.

REV	DESCRIPCION	INGENIERIA			APROBACION	
		Por	Jefe proy.	Fecha	Jefe de proy.	Fecha
0	GR Ingeniería.	Giovani Rojas	Giovani Rojas	25-11-15		
B	Revisión Interna	Ricardo Vega V.	Ricardo Vega V.	25-11-15		
A	Revisión Interna	Marcelo Vargas A.	Marcelo Vargas A.	25-11-15		

Design and technical certification documentation of the ARIA-DMH



The certification was done locally by the company:

GR Ingeniería & Servicios
Av. Jerusalem No. 4166
Antofagasta, Chile
Jv.gringenieria@gmail.com

This is an example of one of the testing performed to the ARIA-DMH.

3.4 Testing the attachment in actual R&I of steering accumulators

Once the attachment is manufactured, we recommend testing it in an actual R&I of an accumulator. This test needs to be conducted in a safe area with adequate controls in place to detect and record issues for correction. The operator should start by manipulating / handling a spare accumulator to get accustomed to the different maneuvers needed while changing an accumulator.
Then, the operator should observe and record issues, and the need for any final adjustments.

3.5 Final adjustments

It is important to make the final adjustments after testing the attachment. In our case, we detected a torsional effect that could destabilize the accumulator when it was lowered to the floor. (2) plates were welded to the base of the hydraulic lift as a resolution - shown in the pictures below.

3.6 Documentation of the standard operating procedure (SOP)

In order to ensure the correct application and operation of this tool, we recommend documenting the Standard Operating Procedure (SOP).
We also recommend reviewing and including any local safety regulations that may apply to the use of this tool.

3.7 Training of the technicians

Before releasing the ARIA-DMH to normal operation, it is important to train the technicians that are going to use it regularly. Consider training at least 2 technicians per shift.

It is recommended to include the application of the SOP in the training, as well as hands on practice using the ARIA-DMH in a controlled and safe environment.

3.8 Release of the ARIA-DMH to work and records of the benefits

Once the ARIA-DMH gets released to operation, we recommend following up closely with accumulator R&I events to ensure the correct use of the tool as well as the documentation of the benefits.

Once the benefits have been established for your operation, we recommend updating the Standard Jobs maintained and applied by the Planning & Scheduling department with the new time and labor (manhours) estimates for this job.

4.0 Benefits

Based on our records and observations, the benefits we have been able to document are represented in the table below.

	Risk level without ARIA-DMH	Risk Level with ARIA - DMH	Number of Technicians assigned w/o ARIA-DMH	Number of Technicians assigned with ARIA-DMH	Total machine down time (DT) w/o ARIA-DMH	Total machine down time with ARIA-DMH	Labor Hours w/o ARIA - DMH	Labor Hours with ARIA-DMH
Steering Accumulator Change	High	Controlled - Low	3	2	7	4	21	8

The results table shows that benefits are observed in the areas of Safety, Execution Time, Use of Labor and Quality of the execution.

Safety: With the use of the ARIA-DMH, the technician will change the work mode from “suspended load” to “supported load”, and will also add a level of control to the whole task. These two factors will reduce the level of risk for this task.

Execution Time (Machine Down Time): The time reduction we recorded is up to 42%, meaning that the total DT of the machine will be 4 hours vs. 7 hours formerly required for this task.

It is important to mention that the reduction of downtime will directly influence the fleet MTTR results. The final influence on the MTTR results will have to be determined based on the frequency of the accumulator change.

Labor Hours (LH utilized): The reduction of one support person means a total reduction of Labor Hours of 13 hours, a 33% reduction, which can be allocated to a different task.
This labor reduction will ultimately influence the results of MR (maintenance ratio) of the equipment fleet positively.

Quality: Although it is difficult to quantify quality, it is evident that by improving safety, the level of control of the task and the technician effort, there will be a significant improvement on the quality of the job done. Any other quality adjustment of the tasks will positively influence the MTBS of the equipment.

5.0 Resources Required

The resources explained and presented in this BP are:

- 1 hydraulic lift as specified
- 1 attachment “ARIA-DMH” that needs to be fabricated.

6.0 Supporting Attachments / References

We attached the following files to this BP:

- Fabrication schematic

7.0 Related Best Practices

None.

8.0 Acknowledgements

This Best Practice was generated, implemented & documented by:

Fernando Valdés
Gerente de Contrato
Contrato Ministro Hales (Codelco Chile)
Telephone: +56 984180500
E-mail: fernando.valdes@finning.com

Marcos Palma Albornoz
Jefe Depto. Mejoramiento Continuo
Contrato Ministro Hales (Codelco Chile)
Telephone: +56 961254692
E-mail: marcos.palma@finning.com

Best Practice written by:

Iván Parra Coloma
Jefe de Mejoramiento de Procesos
Finning SA – Antofagasta
Telephone: +56 990894028
E-mail: ivan.parra@finning.com

Juan Carlos Concha
Coordinador Programa Mejores Prácticas
Finning SA - Antofagasta
Telephone: +56 993096723
E-mail: Juan.Concha@finning.com

Oscar Vejar Saez
MEM Consultant
Caterpillar Global Mining
Telephone: +56 981366251
E-mail: Vejar_Saez_Oscar@cat.com

Abelardo A. Flores
Market Professional - MEM
Caterpillar Global Mining
Retired